Depropanizer Column Ver 1 · OTS

DC3-V1-PM-DOC-001

Document register and engineering queries

Rev 0 · Issued for teaching

Unit	Depropanizer Column Ver 1 · OTS
Document number	DC3-V1-PM-DOC-001
Title	Document register and engineering queries
Discipline	Project management
Originator / DRAWN	Apinan Soottitantawat
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Revision record

REV	DATE	DESCRIPTION	BY	CHK	APP
0	25-OCT-26	Issued for teaching	A. Soottitantawat		

Note. *Numbers come from the simulator model (design case) and the assumptions listed; not for construction. Codes are named for the method only; no client, contractor, licensor or vendor is implied.*

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1 · Purpose, numbering and statuses

This register lists every document of the C-201 depropanizer unit of the OTS: the drawings and studies that existed before the Engineering (EPC) group (PFD, P&ID, HAZOP report, C&E sheet, SRS) and the deliverables generated by the Engineering tabs from the simulator model. It gives the number, title, discipline, revision (read from the PDF cover), status, page count (counted from the file) and file of each, the dependencies between them, the effect of typical design changes, and the register of every engineering query raised while the documents were generated.

Numbering (this project's scheme, assumed)

DC3 unit · V1 version of the unit model · discipline (PM project, PE process, SA safety, ME mechanical, PL plant layout, PI piping, IN instrumentation, EE electrical, PC procurement, CM commissioning) · document type · serial. DC-PFD-200 keeps an older number (EJ-01).

Statuses and roles

- IFR issued for review · IFA issued for approval · IFC issued for construction · IFP issued for purchase · as-built. Letters (A, B …) before approval, numbers (0, 1 …) from IFC (typical practice).
- Originator: discipline engineer; checker: another engineer of the discipline (CHK); approver: lead discipline engineer / engineering manager (APP); document control issues and keeps this register; the client returns a review code (1 no comment, 2 proceed with comments, 3 revise and resubmit, 4 for information).
- Every document of this unit is "Issued for teaching": CHK and APP are blank; none is approved for construction.

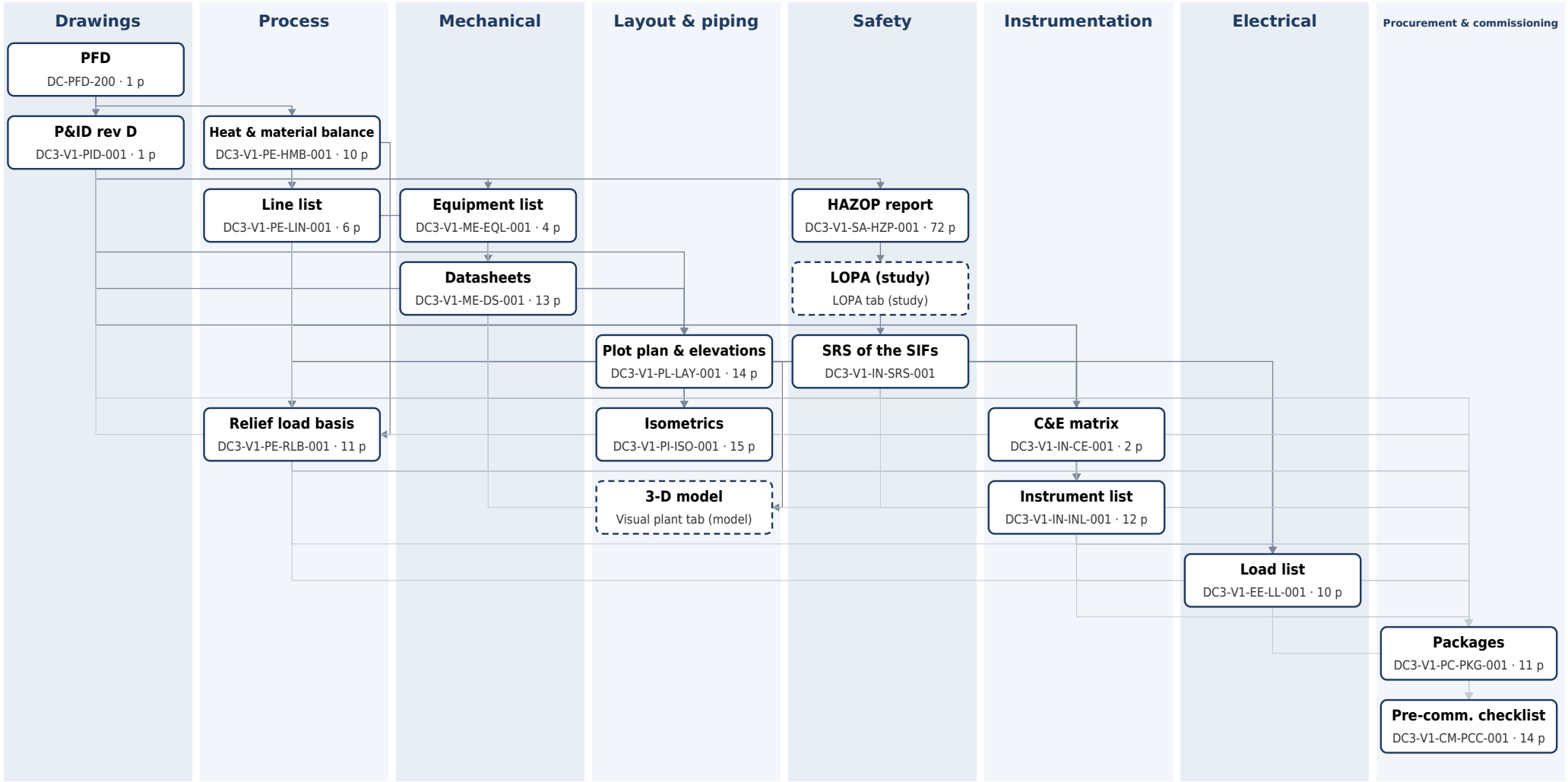
2 · Document register

Document no.	Title	Discipline	Rev	Status	Pages	File	Taught in (tab)	In a real project
DC3-V1-PM-DOC-001	Document register and engineering queries	Project management	0	Issued for teaching	14	DC3-V1-PM-DOC-001.pdf	Project & deliverables	all phases: issued with each transmittal
DC-PFD-200	Process flow diagram, C-201 depropanizer	Process	0	Issued for teaching	1	dc-pfd-200.pdf (+ png, pptx, svg)	PFD	FEED: IFR → IFA → IFC
DC3-V1-PE-HMB-001	Heat and material balance (design, maximum feed, turndown)	Process	0	Issued for teaching	10	DC3-V1-PE-HMB-001.pdf	Process engineering	FEED: IFR → IFA → IFC
DC3-V1-PE-LIN-001	Line list and line sizing	Process	0	Issued for teaching	6	DC3-V1-PE-LIN-001.pdf	Process engineering	FEED draft → IFA → IFC
DC3-V1-PE-RLB-001	Relief load basis	Process	0	Issued for teaching	11	DC3-V1-PE-RLB-001.pdf	Relief & vent	FEED: IFA → IFC
DC3-V1-PID-001	P&ID, C-201 (screen drawing rev C; EPC alternative version rev D)	Process	D	Issued for teaching	1	DC3-V1-PID-001.pdf (+ png, pptx, svg)	P&ID	FEED: IFR → IFA (HAZOP) → IFC → as-built
DC3-V1-SA-HZP-001	HAZOP report	Safety	A	Issued for teaching	72	DC3-V1-SA-HZP-001.pdf	HAZOP	FEED: report on the IFA P&ID, close-out before IFC
DC3-V1-ME-DS-001	Equipment datasheets (C-201, V-201, V-202, E-201, E-202, E-203, P-201A/B, P-202A/B)	Mechanical	0	Issued for teaching	13	DC3-V1-ME-DS-001.pdf	Mechanical	FEED: IFP (enquiry) → IFC / vendor data
DC3-V1-ME-EQL-001	Equipment list	Mechanical	0	Issued for teaching	4	DC3-V1-ME-EQL-001.pdf	Mechanical	FEED: IFA → IFC
DC3-V1-PI-ISO-001	Piping isometrics (main process, steam and relief lines, with line data and BOM)	Layout & piping	0	Issued for teaching	15	DC3-V1-PI-ISO-001.pdf	Layout & isometrics	detailed: IFC per line, then as-built
DC3-V1-PL-LAY-001	Plot plan and elevations (plot plan, C-201 elevation, NPSH sections, spacing and reading levels)	Layout & piping	0	Issued for teaching	14	DC3-V1-PL-LAY-001.pdf	Layout & isometrics	FEED: IFR → IFA (model review) → IFC → as-built
DC3-V1-IN-CE-001	Cause and effect matrix	Instrumentation	R0	Issued for teaching	2	DC3-V1-IN-CE-001.pdf	C&E matrix	detailed: IFA → IFC
DC3-V1-IN-INL-001	Instrument list and I/O count	Instrumentation	0	Issued for teaching	12	DC3-V1-IN-INL-001.pdf	Instrumentation & control	detailed: IFA → IFC → as-built
DC3-V1-IN-SRS-001	Safety requirements specification of the SIFs	Instrumentation	R0	Issued for teaching	—	in the SRS tab, no PDF (EJ-04)	SRS of SIF	detailed: IFA → IFC (frozen before the SIS FAT)
DC3-V1-EE-LL-001	Electrical load list	Electrical	0	Issued for teaching	10	DC3-V1-EE-LL-001.pdf	Electrical	detailed: IFA → IFC
DC3-V1-PC-PKG-001	Procurement package list	Procurement	0	Issued for teaching	11	DC3-V1-PC-PKG-001.pdf	Procurement to commissioning	procurement: IFP, updated per order
DC3-V1-CM-PCC-001	Pre-commissioning checklist	Commissioning	0	Issued for teaching	14	DC3-V1-CM-PCC-001.pdf	Procurement to commissioning	construction: used for MC sign-off
—	Instructor workbook (instructor edition)	Training	—	Issued for teaching	—	instructor edition only; not in the student folder	Instructor station (instructor edition)	—
—	Teaching slide decks (15 files)	Training	—	Issued for teaching	—	one deck per teaching tab, HTML beside the page	—	—

Shaded: generated by the Engineering tabs (12 documents). 16 PDFs, 210 pages in all, counted from the files of the build list (37 files published beside the page). Revision as printed on the PDF cover; where it differs from the draft register in epc/epc.json: DC3-V1-IN-CE-001 cover R0 vs draft 0 (EJ-02).

3 · Deliverable dependencies

Rows are the longest chain from the PFD: a document can only be finished after everything above it. Edges into the package list and the pre-commissioning checklist are the documents those two cite; the others are the inputs each generator reads. A dashed box is a study without a register document; the inputs of the package list and the checklist are drawn light.



Edges

From	To	What flows
PFD (DC-PFD-200)	Heat & material balance (DC3-V1-PE-HMB-001)	stream numbers and the flowsheet
PFD (DC-PFD-200)	P&ID rev D (DC3-V1-PID-001)	equipment and main lines

From	To	What flows
Heat & material balance (DC3-V1-PE-HMB-001)	Equipment list (DC3-V1-ME-EQL-001)	operating P/T and duties → design conditions
Heat & material balance (DC3-V1-PE-HMB-001)	Datasheets (DC3-V1-ME-DS-001)	process data of every sheet
Heat & material balance (DC3-V1-PE-HMB-001)	Line list (DC3-V1-PE-LIN-001)	flow and density of every stream
P&ID rev D (DC3-V1-PID-001)	Line list (DC3-V1-PE-LIN-001)	line numbers, from / to
P&ID rev D (DC3-V1-PID-001)	HAZOP report (DC3-V1-SA-HZP-001)	the drawing studied node by node
Equipment list (DC3-V1-ME-EQL-001)	Datasheets (DC3-V1-ME-DS-001)	tags, sizes, design P/T
Equipment list (DC3-V1-ME-EQL-001)	Relief load basis (DC3-V1-PE-RLB-001)	design pressure / MAWP, wetted areas
Heat & material balance (DC3-V1-PE-HMB-001)	Relief load basis (DC3-V1-PE-RLB-001)	compositions and duties at relief
P&ID rev D (DC3-V1-PID-001)	Relief load basis (DC3-V1-PE-RLB-001)	PSV-201 location and orifice (note 10)
HAZOP report (DC3-V1-SA-HZP-001)	LOPA (study) (LOPA tab (study))	scenarios with causes and consequences
LOPA (study) (LOPA tab (study))	SRS of the SIFs (DC3-V1-IN-SRS-001)	SIFs and the required SIL
SRS of the SIFs (DC3-V1-IN-SRS-001)	C&E matrix (DC3-V1-IN-CE-001)	trip settings, delays, final elements
C&E matrix (DC3-V1-IN-CE-001)	Instrument list (DC3-V1-IN-INL-001)	trip and alarm settings per tag
P&ID rev D (DC3-V1-PID-001)	Instrument list (DC3-V1-IN-INL-001)	every bubble and its line
Line list (DC3-V1-PE-LIN-001)	Instrument list (DC3-V1-IN-INL-001)	line of every instrument, meter sizing
Equipment list (DC3-V1-ME-EQL-001)	Instrument list (DC3-V1-IN-INL-001)	equipment-mounted instruments
Heat & material balance (DC3-V1-PE-HMB-001)	Instrument list (DC3-V1-IN-INL-001)	control-valve cases (Cv)
Relief load basis (DC3-V1-PE-RLB-001)	Instrument list (DC3-V1-IN-INL-001)	PSV-201 size (ER-01)
Instrument list (DC3-V1-IN-INL-001)	Load list (DC3-V1-EE-LL-001)	UPS consumers, analyzers
Equipment list (DC3-V1-ME-EQL-001)	Load list (DC3-V1-EE-LL-001)	pump drivers
Heat & material balance (DC3-V1-PE-HMB-001)	Load list (DC3-V1-EE-LL-001)	pump hydraulics → absorbed power
Equipment list (DC3-V1-ME-EQL-001)	Plot plan & elevations (DC3-V1-PL-LAY-001)	equipment sizes and footprints, design P/T
Datasheets (DC3-V1-ME-DS-001)	Plot plan & elevations (DC3-V1-PL-LAY-001)	C-201 nozzles, platforms and elevations
Line list (DC3-V1-PE-LIN-001)	Plot plan & elevations (DC3-V1-PL-LAY-001)	line sizes and routes from / to
P&ID rev D (DC3-V1-PID-001)	Plot plan & elevations (DC3-V1-PL-LAY-001)	every tag and hand valve to place
Plot plan & elevations (DC3-V1-PL-LAY-001)	Isometrics (DC3-V1-PI-ISO-001)	routes, elevations, supports
Line list (DC3-V1-PE-LIN-001)	Isometrics (DC3-V1-PI-ISO-001)	line number, size, class, design P/T
Plot plan & elevations (DC3-V1-PL-LAY-001)	3-D model (Visual plant tab (model))	layout.json: positions, platforms, ladders, the round
Plot plan & elevations (DC3-V1-PL-LAY-001)	Relief load basis (DC3-V1-PE-RLB-001)	elevations for the fire-case wetted areas (LA-02)
Load list (DC3-V1-EE-LL-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
C&E matrix (DC3-V1-IN-CE-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Instrument list (DC3-V1-IN-INL-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
SRS of the SIFs (DC3-V1-IN-SRS-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Datasheets (DC3-V1-ME-DS-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Equipment list (DC3-V1-ME-EQL-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Heat & material balance (DC3-V1-PE-HMB-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Line list (DC3-V1-PE-LIN-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Relief load basis (DC3-V1-PE-RLB-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
P&ID rev D (DC3-V1-PID-001)	Packages (DC3-V1-PC-PKG-001)	specification / test basis (package docs in DC3-V1-PC-PKG-001)
Load list (DC3-V1-EE-LL-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)
C&E matrix (DC3-V1-IN-CE-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)
Instrument list (DC3-V1-IN-INL-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)
Datasheets (DC3-V1-ME-DS-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)

From	To	What flows
Equipment list (DC3-V1-ME-EQL-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)
Line list (DC3-V1-PE-LIN-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)
Relief load basis (DC3-V1-PE-RLB-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)
P&ID rev D (DC3-V1-PID-001)	Pre-comm. checklist (DC3-V1-CM-PCC-001)	specification / test basis (checklist reference in DC3-V1-CM-PCC-001)

4 · How a design change ripples

For each change the documents that carry the changed value are revised first; every document built from them (section 3) is then re-checked and re-issued, in the order of the dependencies. Pages from the register; the effort index multiplies them by an assumed factor for the phase in which the change is raised (FEED 1, detailed engineering 3, after IFC 10, after purchase orders 30, during construction 100 — a "rule of ten", assumed).

Change	Changes first	Then revised (step)	Docs	Pages	Effort FEED / after PO
PSV-201 orifice L → M (answer to ER-01) note 10 of the P&ID and the PSV-201 row of the INL carry the orifice; the RLB records the decision	P&ID rev D, Relief load basis, Instrument list	Line list (1), HAZOP report (1), Plot plan & elevations (1), Load list (1), Packages (1), Pre-comm. checklist (1), LOPA (study) (2), Isometrics (2), 3-D model (2), SRS of the SIFs (3), C&E matrix (4)	14	168	168 / 5040
Design feed 60 → 70 m³/h the design case of the PFD stream table and of the HMB changes; everything sized from them follows	PFD, Heat & material balance	P&ID rev D (1), Equipment list (1), Line list (1), Datasheets (1), Relief load basis (1), Instrument list (1), Load list (1), Packages (1), HAZOP report (2), Plot plan & elevations (2), Isometrics (2), Pre-comm. checklist (2), LOPA (study) (3), 3-D model (3), SRS of the SIFs (4), C&E matrix (5)	18	196	196 / 5880
PSV-201 set 1.80 → 1.90 MPa g (answer to EP-01) set pressure on the P&ID and in the INL; relieving pressure and loads in the RLB	P&ID rev D, Relief load basis, Instrument list	Line list (1), HAZOP report (1), Plot plan & elevations (1), Load list (1), Packages (1), Pre-comm. checklist (1), LOPA (study) (2), Isometrics (2), 3-D model (2), SRS of the SIFs (3), C&E matrix (4)	14	168	168 / 5040
Add the PSHH-221 trip (EC-07) a new SIF: new transmitter on the P&ID, LOPA credit, SRS entry	P&ID rev D, LOPA (study), SRS of the SIFs	Line list (1), HAZOP report (1), Plot plan & elevations (1), Relief load basis (1), C&E matrix (1), Instrument list (1), Packages (1), Pre-comm. checklist (1), Isometrics (2), 3-D model (2), Load list (2)	14	168	168 / 5040
Line 6"-P-205 → 8" (EP-02) line size in the line list and on the P&ID	P&ID rev D, Line list	HAZOP report (1), Plot plan & elevations (1), Isometrics (1), Relief load basis (1), Instrument list (1), Packages (1), Pre-comm. checklist (1), LOPA (study) (2), 3-D model (2), Load list (2), SRS of the SIFs (3), C&E matrix (4)	14	168	168 / 5040
P-201A motor 75 → 90 kW motor rating in the equipment list, the pump datasheet and the load list	Equipment list, Datasheets, Load list	Plot plan & elevations (1), Relief load basis (1), Instrument list (1), Packages (1), Pre-comm. checklist (1), Isometrics (2), 3-D model (2)	10	104	104 / 3120
Drum elevations from the layout (answer to LA-02) V-201 / V-202 / E-202 elevations on the plot plan and elevations; the fire cases of the RLB re-run	Plot plan & elevations, Relief load basis	Isometrics (1), 3-D model (1), Instrument list (1), Packages (1), Pre-comm. checklist (1), Load list (2)	8	87	87 / 2610

5 · Engineering queries register

60 queries raised while the documents were generated: EP and EM in the shared data (epc/epc.json), ER, EI, EE, EC in each tab's queries.json, EJ by this register. Status OPEN as raised; PARTLY ANSWERED / ANSWERED where a later document answers the query (column "Answered by"); a query is closed only when the documents are revised.

Series	Discipline (tab)	OPEN	PARTLY ANSWERED	ANSWERED	Total
EP	Process (Process engineering)	11	1	—	12
EM	Mechanical (Mechanical)	2	3	1	6
ER	Relief & vent (Relief & vent)	7	—	—	7
EI	Instrumentation (Instrumentation & control)	7	—	—	7
EE	Electrical (Electrical)	4	—	1	5
EC	Procurement to commissioning (Procurement to commissioning)	8	1	—	9
LA	Layout (Layout & isometrics)	9	—	—	9
EJ	Project (Project & deliverables)	5	—	—	5
All		**53**	**5**	**2**	**60**

No.	Raised in	Subject	Query	Proposed answer	Answered by	Status
EP-01	Process engineering	PSV-201 set pressure vs operating pressure	PSV-201 is set at 1.80 MPa g, only 5.9 % above the 1.70 MPa g operating pressure (≥ 10 % is usual; the maximum-feed case runs at 1.702 MPa g). Design-pressure rule (1) gives 1.90 MPa g for C-201 / V-202.	Set PSV-201 at the design pressure 1.90 MPa g (margin 11.8 %) and re-check the relief load and orifice in the Relief tab; or keep 1.80 MPa g and accept simmering risk with a pilot-operated valve.	**DC3-V1-PE-RLB-001** (in part): Relief tab: with the set raised to 1.90 MPa g the governing case still needs an M orifice, so the orifice choice (ER-01) does not depend on EP-01; the set-pressure decision itself is open.	**PARTLY ANSWERED**
EP-02	Process engineering	Line 6"-P-205 outside the sizing criteria	6"-P-205 (Condensate to the reflux drum (gravity)) fails the “Liquid, gravity drain (self-venting)” criteria in the design case / maximum feed: v > 1 (v 1.29 m/s, ΔP 3.99 kPa/100 m); v > 1, ΔP > 5 (v 1.49 m/s, ΔP 5.28 kPa/100 m).	Change to 8" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	—	**OPEN**
EP-03	Process engineering	Line 1"-P-206 outside the sizing criteria	1"-P-206 (Non-condensable vent) fails the “Vapour, vent gas” criteria in the maximum feed: ΔP > 25 (v 10.5 m/s, ΔP 180.33 kPa/100 m).	Change to 1.5" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	—	**OPEN**
EP-04	Process engineering	Line 4"-P-208 outside the sizing criteria	4"-P-208 (Reflux / product pump discharge header) fails the “Liquid, pump discharge / pressure flow” criteria in the maximum feed: v > 3 (v 3.37 m/s, ΔP 43.64 kPa/100 m).	Change to 6" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	—	**OPEN**
EP-05	Process engineering	Line 8"-P-211 outside the sizing criteria	8"-P-211 (Thermosyphon reboiler feed) fails the “Liquid, thermosyphon reboiler inlet” criteria in the design case / maximum feed: v > 1.5, ΔP > 5 (v 1.87 m/s, ΔP 5.84 kPa/100 m); v > 1.5, ΔP > 5 (v 2.15 m/s, ΔP 7.68 kPa/100 m).	Change to 10" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	—	**OPEN**
EP-06	Process engineering	Line 6"-LS-301 outside the sizing criteria	6"-LS-301 (LP steam to the feed preheater) fails the “LP steam” criteria in the design case / maximum feed: v > 30 (v 30.18 m/s, ΔP 13.38 kPa/100 m); v > 30 (v 35.81 m/s, ΔP 18.7 kPa/100 m).	Change to 8" (smallest size meeting the criteria in the maximum-feed case) or justify by a hydraulic check of the actual route; P&ID note 2 says the sizes are preliminary.	—	**OPEN**
EP-07	Process engineering	Condenser E-202 capacity at maximum feed	In the maximum-feed case (70 m³/h) the non-condensables are swept out (blanketing 0 %), PV-210 is at 90.8 % and the vent carries 17.3 kmol/h of hydrocarbon to fuel gas (design 1.3). E-202 has no capacity left.	Treat ≈ 70 m³/h as the hydraulic/thermal limit of the unit in the design basis, or add condenser area (≥ 10 % margin) if the feed is to rise.	—	**OPEN**
EP-08	Process engineering	Product rundown and condensate lines not numbered	The propane rundown downstream of the reflux tee and LV-203, and the steam-condensate lines after ST-201 / ST-202, have no line numbers on DC3-V1-PID-001 rev D.	Number them 3"-P-213, 2"-CD-501 and 2"-CD-502 (sizes from the line list) in the next P&ID revision.	—	**OPEN**
EP-09	Process engineering	Thermosyphon circulation not in the model	The model computes the reboiler boil-up (520 kmol/h) but not the thermosyphon circulation; lines 8"-P-211 and 10"-P-212 were sized with 30 % mass vaporized at the outlet (assumed).	Confirm the circulation ratio with a thermosyphon rating (static head, outlet quality 15–35 %) before freezing the reboiler piping.	—	**OPEN**

No.	Raised in	Subject	Query	Proposed answer	Answered by	Status
EP-10	Process engineering	Volume basis of the flow meters	The model's liquid molar volumes are temperature independent (0.088 / 0.1034 m³/kmol); the HMB m³/h are therefore on a 15 °C basis while the hot bottoms (109 °C) occupy ≈ 22 % more volume.	State the reference temperature of every FI range and use actual density for meter sizing (instrument data sheets).	—	**OPEN**
EP-11	Process engineering	E-201 steam chest pressure at turndown	At turndown (40 m³/h) TV-202 throttles the E-201 shell to 6.7 kPa g (design 148.1): condensate cannot drain through ST-201 against a condensate header at a few hundred kPa g, the shell floods (stall) and the preheat control becomes erratic.	Size ST-201 for the stall condition: pumping trap or condensate pot with level control, or put TV-202 on the condensate side; state the condensate-header back-pressure in the basis.	—	**OPEN**
EP-12	Process engineering	NPSH of P-201 relies on the V-201 pressure	NPSHa of P-201 is 20.2 m with V-201 at 0.40 MPa g (feed bubble pressure 312 kPa g at -1 °C). If the drum liquid were at its bubble point only the static head is left: 3.9 m against NPSHr 3 m + 1 m margin.	Confirm how the V-201 pressure is held above the bubble pressure (upstream pressure / blanketing) or raise V-201 by ≥ 0.2 m; give the pump vendor the bubble-point case.	—	**OPEN**
EM-01	Mechanical	Minimum design metal temperature and material	Depressuring LPG auto-refrigerates to about −42 °C; MDMT −46 °C was applied to C-201, V-201, V-202, E-202 shell, the pumps and the hydrocarbon piping. Plain SA-516-70 is not exempt from impact testing at this MDMT (UCS-66).	Specify SA-516-70N impact tested at −46 °C (and LT piping classes A1L / B1L); confirm the depressuring end temperature in the Relief tab.	**DC3-V1-PE-RLB-001** (answers): Depressuring to 50 kPa g cools the V-202 propane to −43.6 °C, not below the −46 °C MDMT: the MDMT holds; impact-tested material as proposed.	**ANSWERED**
EM-02	Mechanical	V-201 design pressure and overpressure protection	Rule (2) sets V-201 at 1.35 MPa g (feed bubble pressure 1.30 MPa g at 50 °C) while it operates at 0.40 MPa g; the P&ID shows no PSV on V-201 (note 11).	Add a PSV on V-201 set at 1.35 MPa g (fire case and upstream blocked-in); confirm the upstream maximum pressure.	**ER-05** (in part): PSV-202 on V-201 sized (1.5H3, set 1.35 MPa g); whether the overfill case exists still depends on the upstream maximum pressure.	**PARTLY ANSWERED**
EM-03	Mechanical	E-202 tube rupture (cooling-water side)	The CW side of E-202 is designed for 0.70 MPa g, below 10/13 of the shell design pressure (1.46 MPa g): a tube rupture would overpressure the cooling-water system with propane.	Raise the CW side to 1.50 MPa g or provide tube-rupture relief (TSV-216 recommended in the Safety tab) — Relief tab to size.	**DC3-V1-PE-RLB-001** (in part): Tube-rupture load quantified: 5,382 kg/h of propane vapour into the cooling water; the CW-side design or a relief device is still to be decided.	**PARTLY ANSWERED**
EM-04	Mechanical	E-201 / E-203 steam side vs process side	The LP-steam shells (0.70 MPa g) are below 10/13 of the tube-side design pressure (E-201 2.85, E-203 1.95 MPa g): a tube rupture sends hydrocarbon into the steam / condensate system.	Tube-rupture case for both shells in the Relief tab; hydrocarbon detection on the condensate return (recommended addition).	**DC3-V1-PE-RLB-001** (in part): Tube-rupture loads 42,304 kg/h (E-201) and 30,581 kg/h (E-203) of liquid hydrocarbon into the steam shells; detection on the condensate return still open.	**PARTLY ANSWERED**
EM-05	Mechanical	Pump discharge design pressure basis	P-201 discharge items (E-201 tubes, 4"-P-202, 6"-P-203) use the shut-off pressure at the normal suction pressure (2.72 MPa g). If V-201 can reach its design pressure 1.35 MPa g, the shut-off rises to 3.67 MPa g.	Decide whether the maximum suction pressure is the V-201 operating or design pressure; with the latter the discharge piping moves to class 300 at 3.70 MPa g.	—	**OPEN**
EM-06	Mechanical	External pressure, wind and seismic not checked	C-201 (1.7 m × 23.0 m) is checked for internal pressure only. Steam-out and cooling of a blocked-in column can pull vacuum; wind and seismic loads set the bottom-course and skirt thickness of tall columns.	Design C-201, V-201 and V-202 for full vacuum (UG-28) or provide vacuum protection; civil/structural to run wind and seismic checks of C-201 and its skirt.	—	**OPEN**
ER-01	Relief & vent	PSV-201 orifice: L on the P&ID, M required	The governing PSV-201 cases (cooling-water failure, reflux failure and power failure, all ending without reflux) need 2,022 mm² for 45,205 kg/h at 1.98 MPa g, more than the L orifice of DC3-V1-PID-001 note 10 (1841 mm²). The Design tab sized L for the overhead vapour with reflux still running (40,767 kg/h, 1,821 mm²); the model run shows V-202 empties 7.7 min after the cooling water is lost, after which the reflux stops.	Specify PSV-201 as 4M6 (M, 2323 mm²) in the P&ID and the PSV datasheet; or keep L only if a dynamic relief study credits the lower heat input at relieving pressure (611 kmol/h, 1,471 mm², orifice L).	—	**OPEN**
ER-02	Relief & vent	PSV-201 built-up back-pressure with a 6" tail pipe	At the rated capacity of the M orifice (51,939 kg/h) a 6" tail pipe of 30 m equivalent length (assumed) to an atmospheric flare header builds up 353 kPa g = 19.6 % of set (choked at the exit): above the 10 % limit of a conventional valve.	Enlarge the tail pipe to 8" right at the valve outlet (7.6 % of set) or specify a balanced-bellows valve; confirm the superimposed back-pressure of the flare header (the basis assumes atmospheric) and add it to the check.	—	**OPEN**
ER-03	Relief & vent	Model PSV-201 relation larger than the selected orifice	The simulator relation $W_{psv} = K.K_{psv} \times ((P - (set - 20))/100)^{1.5}$ gives 1,601 kmol/h at full lift (1.91 MPa g), 1.78 × the API 520 rated capacity of an L orifice and 1.41 × an M orifice. Model peak pressures are therefore slightly optimistic: with K.Kpsv scaled to the L orifice the cooling-water-failure peak is 1.907 MPa g (model 1.868).	In the next model revision set K.Kpsv = 0.213 (M orifice) so that the simulated PSV flows match the relief load basis; no change to the documents.	—	**OPEN**

No.	Raised in	Subject	Query	Proposed answer	Answered by	Status
ER-04	Relief & vent	Relieving temperatures above the overhead-system design temperature	The no-reflux cases (cooling water, reflux, power failure) relieve the vapour of the feed zone at its dew point 77.7 °C at 1.98 MPa g, above the 75 °C design temperature of 8"-P-204, the E-202 shell and V-202 (rule (a): 45 °C + 28 °C). A fire on C-201 relieves vapour from the boiling bottoms at 124 °C (121 % of set) through 8"-P-204 and the PSV-201 inlet (C-201 itself is designed for 140 °C).	Raise the design temperature of 8"-P-204, the PSV-201 inlet line and body, the E-202 shell and V-202 to 85 °C (relieving temperature + margin); record the fire case as an occasional variation accepted under ASME B31.3 302.2.4 or design the PSV inlet line for 140 °C; the PSV datasheet states both relieving temperatures.	—	**OPEN**
ER-05	Relief & vent	PSV-202 on V-201: liquid relief and upstream pressure	The recommended PSV-202 (set 1.35 MPa g) is sized by the overfill case: 66 m³/h of liquid LPG (383 mm²) against fire 6,810 kg/h (361 mm²); both fit an H orifice. The overfill case exists only if the upstream shut-off pressure exceeds 1.35 MPa g (EM-02).	Confirm the upstream maximum pressure; if overfill is credible route PSV-202 to the flare through a knock-out drum able to take liquid, otherwise size PSV-202 for fire only.	—	**OPEN**
ER-06	Relief & vent	Thermal relief of the E-201 tube side	LPG blocked in the E-201 tubes with steam on the shell expands at 55.6 m³/h at the start (UA × ΔT = 7,058 kW): a G orifice, not a ¾" × 1" thermal valve, and the liquid soon reaches its bubble point.	Add TSV-202 on the E-201 tube outlet (recommended addition) relieving to a closed system, and close XV-202 whenever the feed is isolated (the SIS feed trip already closes XV-201 and XV-202 together).	—	**OPEN**
ER-07	Relief & vent	Depressuring valve for C-201 / V-202	C-201 operates at 1.70 MPa g with about 13.3 t of hydrocarbon in C-201 and V-202. Depressuring to 0.69 MPa g in 15 min under fire needs about 33,025 kg/h on average and 48,217 kg/h at the start, above the PSV-201 governing load.	Recommended: BDV-201 (fail-open, to flare) on the V-202 vapour space, sized in detailed engineering by a dynamic depressuring study; the flare header then takes the depressuring load. Alternative: fire-proofing and water deluge of C-201 skirt and V-202 with a documented risk assessment.	—	**OPEN**
EI-01	Instrumentation & control	Auto-start functions share BPCS transmitters	FSLL-201 (auto-start of P-201B) uses FT-201, the transmitter of FIC-201, and PDSL-230 shares PDT-230 with the DCS indicator PDI-230. The draft list places both functions in the SIS while the pump start outputs (HY-201B, HY-202B) and FT-201 are on the DCS.	Auto-starts of spares are not SIFs and are not credited as independent layers: implement FSLL-201 and PDSL-230 in the DCS with the existing transmitters (re-allocate 2 functions, PDT-230 to the DCS), or give each its own transmitter if they are ever credited in LOPA.	—	**OPEN**
EI-02	Instrumentation & control	LS-202 / LS-203 drawn as switches	The trip sensors are drawn as level switches but the SIS and the screens use a level in % (LSLL-202 at 5 %, LSHH-203 at 92 %).	Specify 4–20 mA level transmitters for the SIS (tags LT-202S / LT-203S, or keep LS with a note) with deviation alarms against LT-202 / LT-203.	—	**OPEN**
EI-03	Instrumentation & control	LV-202 flashing service	The bottoms leave the sump at their bubble point (109.0 °C, Pv 1819.2 kPa a) and LV-202 drops them to the debutanizer at 0.48 MPa g: the valve is choked and flashing in every case (required Cv 11.42 at design). The model's square-root valve law does not choke.	Angle-body valve with hardened trim, outlet expanded to the two-phase line size, located close to the debutanizer inlet; size the downstream line for two-phase flow (Process).	—	**OPEN**
EI-04	Instrumentation & control	LV-203: propane flashes into storage	The propane product leaves V-202 at 39.4 °C with a bubble pressure of 1621.8 kPa a, but the model's storage back-pressure is 0.90 MPa g (1001.3 kPa a): LV-203 is choked and the rundown flashes.	Confirm the storage pressure (or add a product cooler) so the rundown stays liquid; until then size LV-203 for flashing service.	—	**OPEN**
EI-05	Instrumentation & control	PV-210 rangeability	The vent flow varies from 21.4 to 727.9 kg/h between turndown and maximum feed (Cv ratio 22.8:1, choked in every case). A single linear valve of Cv 4 would be 2.5 % open at turndown and 57.1 % at maximum feed.	Use a split-range pair (small + large PV) or an equal-percentage valve with a high rangeability trim; the maximum-feed case also needs the condenser margin (EP-07).	—	**OPEN**
EI-06	Instrumentation & control	FT-217 vent flow meter range	FI-217 is ranged 0–800 Nm³/h but reads 73 Nm³/h at design (9 % of span), 29 at turndown and 438 at maximum feed: an orifice meter (about 4:1 usable turndown) cannot cover this.	Use a high-turndown meter for the vent gas (thermal mass or ultrasonic, 0–500 Nm³/h) or two ranges; the vent flow is the operator's main indication of non-condensables and of condenser blanketing.	—	**OPEN**
EI-07	Instrumentation & control	XV limit switches	XV-201, XV-202 and XV-209 have a closed limit switch (ZSC) only.	Add open limit switches (ZSO) so the SIS can confirm the valves are open before start-up and record partial-stroke tests.	—	**OPEN**
EE-01	Electrical	P-202A / P-202B power supply	Both reflux pumps are motor driven (30 kW each). If they share one MCC bus, a bus fault or a voltage dip stops both and the PDSL-230 auto-start cannot help.	Feed P-202A and P-202B from different bus sections (A / B) of the MCC with a bus-tie; consider automatic re-acceleration after short dips.	—	**OPEN**
EE-02	Electrical	General power failure as a relief case	A site power failure stops the cooling-water pumps, the instrument-air compressors, P-201A and P-202A/B together: E-202 loses its cooling while every valve goes to its fail position (FV-204 and PV-210 open).	Relief & vent: confirm the total power failure case for PSV-201 and the flare header (usually it governs the overhead relief load).	**DC3-V1-PE-RLB-001** (answers): Power failure is relief case 5 of the RLB: it ends without reflux and gives the same governing load as the cooling-water failure (45,205 kg/h, orifice M).	**ANSWERED**

No.	Raised in	Subject	Query	Proposed answer	Answered by	Status
EE-03	Electrical	Non-condensables and the gas group	The model's non-condensables are described as methane / hydrogen; at design they are 59 mol % of the vent gas leaving V-202. Methane keeps the area in group IIA; hydrogen above roughly 30 vol % in the released gas would move the vent and fuel-gas areas to IIB + H2 or IIC.	Process to give the composition of the non-condensables; until then specify IIB T3 for equipment around 1"-P-206, PV-210 and V-202.	—	**OPEN**
EE-04	Electrical	P-201B turbine auto-start without power	The steam-turbine spare is the diverse drive for a power failure, but its trip-and-throttle solenoid, governor and the FSLL-201 logic need power.	Feed the solenoid and governor from UPS-DCS (done in this load list) and design the turbine start valve so that the auto-start works on UPS alone; test the start with the MCC dead during commissioning.	—	**OPEN**
EE-05	Electrical	Cable routes, MCC location and fault level	Cable lengths (P-201A 150 m, P-202A 160 m, P-202B 170 m) are assumed: there is no plot plan with the substation. Short-circuit withstand and protection settings are not checked.	Electrical to confirm the substation location, the 400 V fault level and the protective devices, then re-run the cable sizing (thermal short-circuit check $S \geq I\cdot\sqrt{t} / k$).	—	**OPEN**
EC-01	Procurement to commissioning	P-201A/B run-in on water	Run on water, P-201 develops the same head (m) but 1.82 × the pressure and power of LPG (ρ 548.5 kg/m³): at 60 m³/h it absorbs 90.7 kW against a 75 kW motor, and its shut-off pressure on water is 4.22 MPa g against the 2.85 MPa g design of the casing, E-201 tubes and the 4"-P-202 / 6"-P-203 lines (hydrotest 3.705 MPa g).	No water run-in against the plant system. Run P-201A in on water only on a temporary loop with a full-flow relief set below 2.85 MPa g and the flow limited to 46 m³/h (motor), or commission the pumps directly on LPG after the solo run of the motor.	—	**OPEN**
EC-02	Procurement to commissioning	P-202A/B run-in on water	At 81 m³/h on water P-202 would absorb 43.9 kW (2.12 × the 20.7 kW on reflux at ρ 471.7 kg/m³) against a 30 kW motor; shut-off on water 1.51 MPa g is inside the 2.60 MPa g design.	Limit any water run to 47 m³/h by throttling the discharge, with the motor current watched; preferably commission on propane.	—	**OPEN**
EC-03	Procurement to commissioning	C-201 hydrotest weight for the foundation and skirt	Full of water C-201 weighs 76.6 t (empty 23.2 t + 53.4 m³ water) against about 27.6 t in operation (model holdup 4.4 t). A vertical field hydrotest also raises the bottom to 2.704 MPa g (water head 0.234 MPa over 23.82 m).	Design the foundation, skirt and anchor bolts for the hydrotest load case; test C-201 in the shop horizontally at 2.535 MPa g and in the field only after a repair, with civil confirmation.	—	**OPEN**
EC-04	Procurement to commissioning	Tightness test pressure	The N_2 tightness test is limited to 0.80 MPa g by the N_2 header (assumed); a test at the operating pressure (up to 1.740 MPa g in the column system) would also exceed 90 % of the PSV-201 set (1.62 MPa g) because the set is only 5.9 % above operating (EP-01).	N_2 test at the header pressure; the range above it is proven by a hydrocarbon leak check at each pressure step of the start-up (hot bolting rounds). Revisit when EP-01 (PSV set) is answered.	**DC3-V1-PE-RLB-001** (in part): Depends on EP-01 (PSV set); the RLB keeps 1.80 MPa g, so the 0.80 MPa g N_2 test stands until EP-01 is decided.	**PARTLY ANSWERED**
EC-05	Procurement to commissioning	Long-lead items on the critical path	With the assumed lead times PKG-01 Column C-201 (44 weeks, float 2 weeks) and PKG-04 Pumps P-201A/B, P-202A/B (48 weeks, float 2 weeks) have ≤ 4 weeks of float to their construction need dates; one month of delay in their purchase orders delays mechanical completion.	Place the column and pump purchase orders on the FEED datasheets (DC3-V1-ME-DS-001) with a later technical freeze, expedite monthly, and plan the column erection as the first heavy lift.	—	**OPEN**
EC-06	Procurement to commissioning	Dryness criterion after the hydrotest	The design basis has no dryness criterion; −20 °C dew point is assumed. The feed arrives at −1 °C and propane auto-refrigerates to −42 °C on depressuring: free water would form ice / hydrates in valves and instrument taps.	Confirm the dew point (−20 °C or lower) with the propane product water specification; dry with dry N_2 after the hydrotest.	—	**OPEN**
EC-07	Procurement to commissioning	PSHH-221 in the SIS FAT scope	PSHH-221 is a recommended SIF (HAZOP, LOPA, SRS) but not in the logic; the SIS FAT tests 6 causes. Adding it after the FAT means a second FAT and re-validation.	Decide on PSHH-221 (PT-221, own nozzle) before the SIS configuration freeze so it is tested at the FAT.	—	**OPEN**
EC-08	Procurement to commissioning	Performance-test corrections and tolerances	The guarantees are stated at the design case (cooling water 330 m³/h at 15 °C, LP steam 0.45 MPa g, feed z = 0.5707); on the test day these differ. Tolerances assumed: capacity −1 %, steam +5 %.	Agree in the contract the correction curves (CW temperature, feed composition) and the measurement uncertainty; calibrate FI-201, FI-205, FI-206 and the analyzers before the test.	—	**OPEN**
EC-09	Procurement to commissioning	OTS start-up begins under fuel gas	The simulator's start-up procedure starts from the cold state under a fuel-gas blanket: N_2 purging, gassing-up and the oxygen check of SOP-C201-01 are not in the simulator.	Keep them as field steps in the commissioning procedure; brief the operators on the OTS for the steps after gassing-up only.	—	**OPEN**
LA-01	Layout & isometrics	Relief line from PSV-201 to the flare header not numbered	The EPC P&ID shows the PSV-201 outlet only as "4"x6"" to flare; the layout routes it as a 41.8 m tail pipe from the top platform down the column to the flare header on the rack top tier, and isometric sheet 5 needs a line number.	Number it 6"-FL-601 (class A1L, design 0.50 MPa g / 140 °C, MDMT −46 °C, assumed) and add it to the line list; if ER-02 is answered with an 8" tail pipe, the number becomes 8"-FL-601.	—	**OPEN**
LA-02	Layout & isometrics	Relief fire-case elevations differ from the layout	The relief basis assumed V-201 bottom at EL +1.0, V-202 bottom at EL +1.5 and E-202 bottom at EL +8.0. The layout puts both drums at EL +5.2 (NPSH with the liquid at its bubble point) and E-202 at EL +8.6. Both drums remain inside the 7.6 m fire zone but the wetted area below 7.6 m changes.	Re-run the fire cases of DC3-V1-PE-RLB-001 with the layout elevations (V-201 / V-202 wetted area up to EL +7.6); keep the drum elevations, which also answer EP-12 (NPSH of P-201 without relying on the V-201 pressure).	—	**OPEN**

No.	Raised in	Subject	Query	Proposed answer	Answered by	Status
LA-03	Layout & isometrics	Model static head of the pump suctions differs from the layout	The simulator uses a P-201 suction static head K.stat1 = 23 kPa and the pump hydraulics 30 kPa for P-202. With the layout elevations the static head at the normal level is 29.1 kPa (V-201, 5.41 m) and 24.2 kPa (V-202, 5.22 m).	Update K.stat1 to 29 kPa and the P-202 value to 24 kPa at the next model revision (raises NPSHa by 1.1 m; no effect on the exercises), or record the difference in the model assumptions.	—	**OPEN**
LA-04	Layout & isometrics	TI-201 drawn in the V-201 vapour space	On the EPC P&ID the TI-201 bubble is connected to the top of V-201. A thermowell in the vapour space reads the vapour temperature, not the feed liquid the pumps take; the instrument list puts TT-201 on 6"-P-201.	Place the TI-201 thermowell in the V-201 liquid space (layout: 0.5 m above the bottom on the south side, read from the S-202 table top) or on 6"-P-201, and correct the P&ID at the next revision.	—	**OPEN**
LA-05	Layout & isometrics	TI-213 not within reach (reading-level check)	TI-213 at EL +7.40 is 2.80 m above S-201-1: the 6"-P-205 drop from E-202 to V-202 has no floor below it within 2.0 m.	Move the TI-213 thermowell to the V-202 outlet 6"-P-207 near the pumps (same subcooled liquid, read from grade), or add a local platform on the V-202 top; the DCS reading is not affected.	—	**OPEN**
LA-06	Layout & isometrics	Vents and drains unnumbered on the P&ID	The V-201 top vent, the V-201 bottom drain and the 4"-P-210 low-point drain are drawn as "VENT" and "DR" without tag numbers; the layout and the 3-D model need a tag for each hand valve (VENT-V201, DR-V201, DR-P210 used here, marked recommended).	Give every vent and drain valve a number in the valve list (e.g. HV-2xx) and show it on the P&ID; keep them normally closed and plugged.	—	**OPEN**
LA-07	Layout & isometrics	PSV-201 tail-pipe length longer than the relief-basis assumption	ER-02 computed the built-up back-pressure with a 30 m equivalent length. The routed tail pipe is 41.8 m with 4 elbows, about 54 m equivalent (20 D per elbow, assumed), so the built-up back-pressure will be higher than in ER-02.	Recompute ER-02 with the layout length; consider an 8" tail pipe from the valve outlet, a balanced-bellows valve, or mounting PSV-201 lower on the overhead line (it must stay free-draining to the header).	—	**OPEN**
LA-08	Layout & isometrics	Building siting and flare location assumed	The occupied control room is placed 63 m and the substation 57 m from the nearest LPG item (P-201A, P-201A), upwind for a south-west prevailing wind; the flare knock-out drum and the flare are off-plot to the west. None of this is based on a study.	Carry out the building siting study (explosion overpressure and toxic / fire exposure, API RP 752 / 753 practice) and the flare radiation study (API 521) before freezing the plot plan; confirm the prevailing wind from site data.	—	**OPEN**
LA-09	Layout & isometrics	C-201 erection lift and the crane hardstand	C-201 weighs about 23.2 t empty (EQL estimate) plus platforms, ladders, insulation and dressing; it would be lifted from hardstand HS-1, 11.3 m from the column centreline, to a height of 26.4 m.	Select the crane and tailing crane with the dressed weight, confirm the hardstand bearing capacity with civil, and check the lift radius; dress the column (platforms, insulation) at grade where possible.	—	**OPEN**
EJ-01	Project & deliverables	Document numbers outside the project scheme	DC-PFD-200 does not follow the scheme DC3-V1-<discipline>-<type>-<serial> used by the other documents (the PFD was numbered before the scheme existed).	Re-number at the next revision (e.g. DC3-V1-PE-PFD-001) and keep DC-PFD-200 as a superseded number in this register so old references still resolve.	—	**OPEN**
EJ-02	Project & deliverables	Revision notation differs between documents	Revisions in use: 0, A, D, R0. Register draft and PDF cover disagree: DC3-V1-IN-CE-001 (register draft 0, PDF cover R0). Letters are used for the HAZOP report (A) and the P&ID (D), numbers for the generated documents (0), "R0" for the C&E sheet and the SRS.	Adopt one convention for the project: letters (A, B, …) for issues before approval (IFR / IFA), numbers (0, 1, …) from issued for construction; record the cover revision in this register (done: the register shows the PDF cover revision).	—	**OPEN**
EJ-03	Project & deliverables	Two versions of the P&ID in use	The OTS screens show the P&ID rev C; the EPC drawing DC3-V1-PID-001 is rev D (alternative version). The HAZOP report lists both; every generated document refers to rev D.	Register rev D as the controlled drawing; mark rev C "OTS screen drawing, for training only" on the screen and in the decks.	—	**OPEN**
EJ-04	Project & deliverables	SRS has no controlled PDF	DC3-V1-IN-SRS-001 exists only as the SRS tab; the package list (PKG-06, PKG-12) cites it as a specification document for the SIS vendor, who needs a fixed, revisioned copy.	Issue the SRS as a PDF with a revision record (like the C&E sheet) before the SIS enquiry.	—	**OPEN**
EJ-05	Project & deliverables	Hold on PSV-201 data until ER-01 is closed	PSV-201 appears as L on the P&ID rev D, in the instrument list and on the C-201 datasheet, and as M in the relief load basis (ER-01). A purchase order placed now (PKG-07) would buy the wrong valve.	Put a HOLD on PSV-201 in the P&ID, INL and PKG-07 until ER-01 is answered; then revise the documents in the order of part 4 (P&ID → INL → packages).	—	**OPEN**

6 · Cross-document consistency

Earlier statements that later documents answer, supersede or contradict. The earlier value stays until its document is formally revised, with a note pointing to the query.

Earlier statement	Where	Later document / query	Relation	How it was handled
Design tab: first PSV-201 estimate, orifice L (1,821 mm², overhead vapour with the reflux still running)	Design of the unit operation	ER-01	superseded	The relief load basis takes the no-reflux cases: 45,205 kg/h → 2,022 mm² → orifice M (4M6). The P&ID rev D (note 10), the INL and the datasheet keep L until the P&ID is revised and say so.
Equipment list rule (a): overhead system (8"-P-204, E-202 shell, V-202) designed for 75 °C	Mechanical	ER-04	contradicted	Relieving temperature 77.7 °C without reflux and 124 °C in a fire on C-201: above 75 °C. The EQL keeps 75 °C until ER-04 is answered; the datasheet points to ER-04.
Draft instrument list: PSV-201 "API L orifice from the Design tab"	Instrumentation & control	ER-01	noted	The refined INL keeps L (as on the P&ID rev D) and adds "M required by DC3-V1-PE-RLB-001, see ER-01".
Draft instrument list and load list in epc/epc.json used by the package list and the checklist	Procurement to commissioning	DC3-V1-IN-INL-001 / DC3-V1-EE-LL-001	superseded	The procurement generator now reads the refined lists of the Instrumentation and Electrical tabs (109 base INL rows, 11 load-list rows); I/O count identical to the INL.
Relief load basis: fire-case wetted areas from assumed elevations of V-201, V-202 and E-202	Relief & vent	LA-02	contradicted	The relief basis assumed V-201 bottom at EL +1.0, V-202 bottom at EL +1.5 and E-202 bottom at EL +8.0. The layout puts both drums at EL +5.2 (NPSH with the liquid at its bubble point) and E-202 at EL +8.6. Both drums remain inside the 7.6 m fire zone but the wetted area below 7.6 m changes. The RLB keeps its elevations until LA-02 is answered.

The same number in several documents

Number	Documents	First value	Second value	Result	Note
PSV-201 orifice ≥ required area	P&ID rev D · RLB	L 1841 mm²	2,022 mm² → M	✗	ER-01
INL PSV-201 row points to ER-01	INL · RLB	Pressure safety valve 4"×6", orifice L on the P&ID rev D (1841 mm²); M (2323 mm²) required by DC3-V1-PE-RLB-001, see ER-01	mentions ER-01	✓	
PSV-201 set ≤ design pressure of C-201	EQL · model	1.80 MPa g	1.90 MPa g	✓	
PSV-201 set ≤ design pressure of V-202	EQL · model	1.80 MPa g	1.90 MPa g	✓	
PSV-201 set ≤ design pressure of E-202	EQL · model	1.80 MPa g	1.90 MPa g	✓	
Relieving T (no reflux) ≤ overhead design T	RLB · EQL · LIN	77.7 °C	75 °C	✗	ER-04
Relieving T (fire on C-201) ≤ 8"-P-204 design T	RLB · LIN	124 °C	75 °C	✗	ER-04
P-201A motor: HMB = EQL = LL = PKG	HMB · EQL · LL · PKG	75 kW / 75 kW / 75 kW / 75 kW	all equal	✓	
P-202A motor: HMB = EQL = LL = PKG	HMB · EQL · LL · PKG	30 kW / 30 kW / 30 kW / 30 kW	all equal	✓	
I/O count: INL = FAT scope of the package list	INL · PKG	DCS AI 31, DCS AO 7, DCS DI 7, DCS DO 4, SIS AI 3, SIS DI 5, SIS DO 3 = 60	60	✓	
INL rows / load rows / lines in the package list	INL · LL · LIN · PKG	109 / 11 / 20	109 / 11 / 20	✓	

7 · Assumptions

- Document numbering scheme, the status path and the roles are typical EPC practice, assumed for teaching.
- Page counts and cover revisions are read from the PDF files of the build list (student folder and epc/pdf); the instructor workbook (instructor edition) and the teaching decks are not paginated here.
- Dependencies: the inputs each generator of the Engineering tabs reads, plus the documents cited by the package list and the checklist (Procurement to commissioning tab).
- Effort factors 1 / 3 / 10 / 30 / 100 by phase ("rule of ten") are assumed and only rank the changes.
- "Answered by" records where a later document answers an earlier query with its own data; the query stays OPEN in its source file until the documents are revised.